

Frequency:	437.5	MHz		
Orbital Altitude	500	km		
Elevation Angle	35	Deg		
Distance:	815.089	km		
	-10	dBW		
Spacecraft	2.2	dB		
Antenna Gain				
Spacecraft Line	1	dB		
Loss				
EIRP	-8.8			
	GFSK	CW Morse	RTTY	
	(2K00	(500H	FSK	
	F1D)	A1A)	(515H	
			F1B)	
Free Space Path	143.4737	143.47	143.47	dB
Loss:		37	37	
Antenna Pointing	1	1	1	dB
Loss:				
Antenna	1	1	1	dB
Polarization				
Loss:				
Atmospheric	2	2	2	dB
Loss:				
Ionospheric Loss:	1	1	1	dB
Rain Loss:	0	0	0	dB
Isotropic Signal	-157.274	-	-	dB
Level at GS		157.27	157.27	
		4	4	
GS Antenna	0.2	0.2	0.2	dB
Pointing Loss				
GS Antenna Gain	14	14	14	dB
GS Transmission	2	2	2	dB
Loss				
Ground Station	635	635	635	K
Effective Noise				
Temperature:				
Ground Station	-16.2277	-	-	dB/K
Figure of Merit		16.227	16.227	
(G/T):		7	7	
Signal Power at	-145.474	-	-	dBW

Input of LNA		145.47	145.47	
		4	4	
Boltzman's Constant	-228.6	-228.6	-228.6	dbW/K/Hz
Receiver Bandwidth	2000	500	515	Hz
Receiver Noise Power	-167.562	-	-	dBW
		173.58	173.45	
		3	4	
SNR at Receiver	22.0883	28.108	27.980	dB
		9	53	
Required SNR	12.8	10	10	dB
Link Margin	9.288301	18.108	17.980	dB
		9	5	